Dual-Track Execution Plan

Track 1 — a method benchmark that can win on its own. Track 2 — the validation collaboration that is the only path to Nature.

Prepared for H. S. Heo. Companion to the KRAS/Tau scaffolds and the formal novelty report.

How the two tracks relate: Track 1 is self-contained and publishable in a computational venue if the DEQ shows a measured advantage; it also de-risks the method and becomes the credibility 'calling card' that makes Track 2 outreach far more compelling. Track 2 is the only axis the field reliably rewards with a top-tier outcome, but it depends on a wet-lab partner. Run Track 1 now; open Track 2 outreach in parallel, leading the pitch with Track 1 progress.

TRACK 1 — Method benchmark: DEQ vs. incumbents

1.1 The single hypothesis under test

H1: a deep-equilibrium (implicit fixed-point) formulation predicts a self-consistent, condition-dependent phase-separation observable better than feed-forward, equilibrium-distribution (DiG-style), and active-learning baselines — specifically in the regime that requires extrapolation to unseen conditions, where an equilibrium inductive bias should pay off. If H1 is false, we report it; we do not retune until a win appears.

1.2 Task and public datasets

Primary task: predict the condition-dependent phase boundary / saturation concentration (and a binary LLPS yes/no) for IDP sequences under specified conditions. This task is chosen because it is exactly where a self-consistent equilibrium representation is most defensible, and where strong public baselines exist.

- Data: LLPSDB v2 (condition-annotated in-vitro LLPS), PhaSePred / DrLLPS / CD-CODE (condensate components and propensity), and curated saturation-concentration sets where available. All records keep source IDs (traceability).
- Splits: by protein family AND by condition regime (e.g., train low-salt, test high-salt) with a temporal embargo where dates exist. Leakage control is the point of the benchmark, not an afterthought.

1.3 Baselines and ablations (all re-implemented on the same splits)

Model	What it represents	Role in the comparison
ESM-2 embedding + MLP	Strong feed-forward sequence baseline	Lower bound; tests whether equilibrium structure helps at all
Condition-aware ML (LLPSense-style)	PLM embeddings + environmental variables	Current SOTA for condition-dependent LLPS
DiG-style equilibrium-distribution model	Learns an equilibrium distribution; bias for design	The closest published 'equilibrium + inverse design' competitor
Active-ML phase-diagram navigator	Iterative experiment/sample selection	Tests whether DEQ beats smart sampling, not just static models
catGRANULE / FuzDrop	Heuristic propensity predictors	Weak baseline / sanity floor

Model	What it represents	Role in the comparison
DEQ (candidate)	Implicit fixed-point, O(1)-memory, differentiate-through-equilibrium	The method under test
Ablation A: DEQ -> unrolled finite depth	Removes implicit differentiation	Isolates the value of the implicit-function gradient
Ablation B: DEQ -> no condition coupling	Removes the self-consistent condition dependence	Isolates the value of the equilibrium inductive bias

1.4 Metrics — with one decisive metric

- In-distribution: AUROC/AUPRC (LLPS yes/no); Spearman rho and RMSE (saturation concentration).

Decisive metric (extrapolation gap): performance on held-out condition regimes. H1 lives or dies here. Report the drop from in-distribution to out-of-condition for every model; the DEQ's claim is a smaller drop.

- Efficiency: peak VRAM vs effective depth (the O(1)-memory advantage), and wall-clock on a single RTX 4090.

1.5 Pre-registered win condition (frozen before runs)

DEQ is declared advantageous only if ALL of: (i) it beats the best baseline on the extrapolation metric by Spearman >= 0.05 with non-overlapping 95% bootstrap CIs across >= 5 seeds; (ii) it matches or beats the best baseline in-distribution; (iii) the memory advantage is quantified. If it wins only on extrapolation + memory but ties in-distribution, that is a legitimate, reportable story. If it loses across the board, the honest output is a null-result note or a preprint — not a re-tune.

1.6 Inverse-design sub-benchmark (connects to the original claim)

Because the contribution invokes inverse design, add a retrospective design test: with each model frozen, gradient-design (DEQ) / bias-sample (DiG) / propose-and-rank (active-ML) sequence mutations predicted to lower the saturation concentration, then score enrichment against held-out, experimentally known LLPS-promoting mutations (e.g., FUS/hnRNPA1/tau disease mutants). Metric: EF@k / BEDROC. This is still retrospective (no wet lab), so it stays honest, but it directly tests 'can the DEQ design, not just predict'.

1.7 Compute, integrity, timeline

- Feasible on a single RTX 4090 (the DEQ memory profile is an asset here). Antigravity executes; author audits, reusing the pre-registration + provenance protocol already written (freeze splits/metrics, leakage control, report null honestly, single-source-of-truth for any derived number).
- Rough timeline: data + splits (3-4 wk), baselines (4-6 wk), DEQ + ablations (4 wk), inverse-design sub-benchmark (3 wk), write-up (4 wk). ~4-5 months solo with Antigravity.
- Target venues if H1 holds: Nature Computational Science / Nature Communications / Journal of Chemical Information and Modeling / Bioinformatics.

TRACK 2 — Validation collaboration (the Nature path)

2.1 You actually need TWO things, not one

A computationally designed molecule cannot validate itself. Reaching a wet-lab readout requires (a) synthesis — someone to make the designed compound (a medicinal-chemistry collaborator or a CRO), and (b) assay — someone to run the tau condensate / KRAS readout. Plan for both; the synthesis need is the one most often forgotten.

2.2 Minimal assay package to request

- Tau (liquid-to-solid): ThT aggregation kinetics with and without seeding; droplet-fusion aspect-ratio or FRAP for material state; on 1-2 designed candidates + an inactive control + a known reference (e.g., a published tau-transition modulator). Established system: P301L tau + polyU RNA, monitored by TIRF/ThT.
- KRAS (cell): biochemical or cell-based target-engagement assay + viability in a KRAS G12C NSCLC line (e.g., NCI-H358), against single-agent and inactive controls.
- Keep it minimal: one decisive, falsifiable prediction confirmed is worth more than a broad screen.

2.3 Candidate collaborators (verify current PI / availability)

Group / institution	Demonstrated capability	Fit / note
Mukhopadhyay group, IISER Mohali (India)	Tau LLPS and chaperone-regulated liquid-to-solid transition; ThT/FRAP (Sci. Adv. 2025)	Closest match to the exact tau L-to-S readout
Ferreon lab, Baylor College of Medicine (USA)	Tau LLPS, oligomers, structural-state vs phase behaviour	Strong tau-condensate biophysics
Mondal group, TIFR Hyderabad (India)	Computational tau condensate / cation modulation, rheology	Method cross-check + possible joint comp/exp
Wei group, Fudan (China)	Computational condensate aging, beta-sheet motifs	Computational corroboration partner
KBRI — Korea Brain Research Institute, Daegu (KR)	Neurodegeneration / tau pathology (institutional)	Geographically close; identify current tau PI
KRIBB, Daejeon (KR)	Protein bioscience, biochemistry (institutional)	Same city as author; identify relevant group
KAIST / SNU / IBS centres (KR)	Condensate imaging, synthetic condensates, biophysics	Via advisor / Dr. An network; identify current PI

Start with the advisor's and Dr. An's networks for a warm introduction to a Korean group; use the international groups as the gold-standard fallback. Verify each group is currently active on this assay before outreach.

2.4 Outreach pitch (template — fill the brackets)

"Dear Prof. [name], I am a doctoral researcher working on a deep-equilibrium computational framework for modelling ligand-stabilised conformational states relevant to the tau liquid-to-solid

transition. I have a small number of specific, falsifiable predictions for molecules that the model expects to [slow/accelerate] the transition, and I am looking for a wet-lab collaborator to test 1-2 of them with a ThT/droplet-fusion readout. I would handle all computational work and pre-register the predictions; I am proposing a genuine co-authorship collaboration, not a service request. Might I send a one-page concept with the specific predictions? Best regards, [name, affiliation]."

What you bring / what you need. Bring: the design, the Antigravity pipeline, pre-registered falsifiable predictions, the integrity protocol, and full computational write-up. Need: synthesis of 1-2 compounds and the assay. Be explicit and modest about the ask — one decisive test, not an open-ended program.

2.5 Terms to settle up front (before data exists)

- Authorship: agree contribution and order in writing before experiments (avoids the disputes that derail collaborations).
- Pre-registration: the predictions are timestamped before the assay, so a confirmation is credibly prospective.
- Data/IP: clarify ownership and any SimsReality / institutional obligations early, given the author's existing commitments.
- Honest framing: 'validated hypothesis-generation framework', not 'discovered therapy' — the same standard as the scaffolds.

Sequencing and milestones

1. Now: freeze Track-1 pre-registration (task, splits, metrics, win condition). Begin data assembly.
2. Parallel: shortlist 3 Track-2 groups; ask advisor / Dr. An for a warm intro to one Korean group.
3. Month 2-4: run Track-1 baselines + DEQ + ablations; draft the one-page Track-2 concept with specific predictions once the model is trained.
4. Month 4-5: Track-1 write-up (submit if H1 holds). Send Track-2 outreach, leading with Track-1 results as the calling card.
5. Month 5+: if a partner commits, run the minimal validation; only then assemble the higher-tier biomedical manuscript.

References (verify every entry before use)

Recovered from web-search passes; confirm authors, venue, year, DOI against the primary source.

1. Rai, S. K. et al. (Mukhopadhyay group). Chaperone-mediated heterotypic phase separation regulates liquid-to-solid transition of tau into amyloid fibrils. Sci. Adv. 11, eads1241 (2025). [verify]

2. Lucas, L., Tsoi, P. S., Ferreon, J. C. & Ferreon, A. C. M. Tau oligomers resist phase separation. Biomolecules 15, 336 (2025). [verify]

3. [Tau aggregation inhibitor methylene blue accelerates liquid-to-gel transition while inhibiting fibrils.] (2023). [verify]

4. [Liquid-solid coexistence at single-fibril resolution during tau condensate ageing; P301L tau + polyU; TIRF/ThT.] bioRxiv (2025). [verify]

5. LLPSense: ML framework for predicting and manipulating condition-dependent protein phase separation. bioRxiv (2025). [verify — Track-1 baseline]
6. PhaSeMotif: interpretable & generative DL for phase-separating IDR motifs (validated). Nat. Commun. (2026). [verify — Track-1 baseline]
7. [DiG] Predicting equilibrium distributions for molecular systems with deep learning. Nat. Mach. Intell. (2024). [verify — Track-1 baseline]
8. Bai, S., Kolter, J. Z. & Koltun, V. Deep equilibrium models. NeurIPS 32 (2019).
9. LLPSDB / PhaSePred / DrLLPS / CD-CODE — public LLPS datasets for Track-1 (cite specific releases). [verify]

— End of plan —